

Grad Cafe Analytics Summary

1. Average GRE score of accepted PhD applicants

The average GRE score of accepted PhD applicants was 266.8 based on 608 applicants with available GRE scores.

2. Average GPA of accepted Master's applicants

The average GPA of accepted Master's applicants was 3.70 based on 4598 applicants with valid GPA values.

3. Shortcomings of inferring decision_date

Inferring decision dates from status text can be inaccurate because entries are user-written and may contain missing or inconsistent dates. The term value can also cause incorrect year assignments when decisions are posted outside the expected timeline. 168 rows had a computed days_to_decision that was negative or unrealistically large (over 365 days) and were excluded from downstream analysis.

4. Correlation analysis

The correlation analysis showed that GPA and GRE scores had a weak negative relationship ($r = -0.016$). This suggests academic performance and GRE scores are related to some degree, but one measure cannot reliably predict the other on its own.

5. Accepted vs rejected GPA comparison

Accepted (mean GPA = 3.751, $n=8467$) and rejected (mean GPA = 3.772, $n=9275$) applicants had different GPA averages. A Mann-Whitney U test showed that GPA was significantly different between groups ($U = 37872086.0$, $p = 4.189e-05$).

6. Chi-square test

The chi-square test showed that there was a significant relationship between degree type and applicant origin (chi-square = 813.002, $p = 1.172e-174$, $dof = 4$).

7. Remaining data quality issues

Remaining issues include missing values, inconsistent text fields, possible duplicates, and self-reported data bias. GPA and GRE AW values of exactly zero were treated as missing rather than real scores, and GRE values were accepted under either the modern single-section scale or the older combined scale, since the raw data mixed both without indicating which was used. These limitations mean the dataset may not fully represent all graduate applicants.